



Tehran University of Medical Sciences

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Final Exam
International Campus
Cell & Molecules, Biochemistry (practical)
MD, DDS, PharmaD Program
September 2023 semester

Name and Surname:

Student ID Number:

Course Coordinator: Dr. Dashti

Date of Exam: Jan. 22, 2024

Duration of Exam: 25 minutes

MCQ Question

25

Directions:

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil.

1. In the presence of concentrated sulfuric acid, carbohydrates lose.....and form.....?

- a) CO_2 , diazine
- b) H_2O , furfural
- c) H_2O , diazine
- d) O_2 , furfural

2. The reagent used for distinguishing a reducing monosaccharide from a reducing disaccharide is ...

- a) Barfoed's reagent
- b) Fehling's reagent
- c) Selwinoff's reagent
- d) Benedict's reagent

3. Which of the following test can be used as a semi quantitative test for sugars?

- a) Molisch test
- b) Seliwanoff test
- c) Benedict test
- d) Barfoed test

4. If an amino acid solution has a positive result for the Hopkins-cole's test, so:

- a) It has a guanidino group
- b) It has a pyrrol ring
- c) It has an imidazole group
- d) It has an indole ring

5. If we add trichloro acetic acid to an unknown solution, we will observe a precipitation so:

- a) The solution is monosaccharide
- b) The solution is cholesterol
- c) The solution is carbohydrate
- d) The solution is protein

6. Which of the following tests are used for detection of histidine?

- a) Xanthoproteic acid test
- b) Biuret test
- c) Pauly's diazo test
- d) Sakaguchi test



7. which part of spectrophotometer machine convert radiant energy (photons) into an electrical signal?

- a) Light source
- b) Detector ✓
- c) Monochromator
- d) Cuvettes

8. In all of the following conditions cause hyperproteinemia, except:

- a) Persistent vomiting
- b) Chronic inflammation. ✓
- c) Viral infections ✓
- d) Hepatic failure ✓

9. Which of the following is not in the Beer's Law equation?

- a) Molar absorptivity
- b) Cell path length ✓
- c) Light wavelength
- d) Concentration

★ 10. The product of urease reaction is and for detection of this product we use

- a) NH_3 , phenol red
- b) H_2O , phenolphthalein ✓
- c) H_2O , phenol red
- d) CO_2 , phenolphthalein

11. Which of the following is correct about "the effect of temperature on reaction rate" experiment:

- a) Renin activity is constant at the different temperature. ✓
- b) Renin has a maximum activity at 80°C . ✓
- c) Renin has a maximum activity at 37°C . ✓
- d) At the lower temperatures, the rate of reaction will be increased. ✓

12. The optimum pH for the action of pepsin is and salivary amylase is

- a) acidic- neutral ✓
- b) basic- acidic
- c) neutral-basic
- d) basic- neutral

13. In chromatography, which of the following can the mobile phase be made of?

- a) Solid or liquid
- ☒ b) Liquid or gas
- c) Gas only
- d) Liquid only

14. Chromatography is a physical method that is used to separate and analyse _____

- a) Simple mixtures
- ☒ b) Complex mixtures
- c) Viscous mixtures
- d) Metals

15. What happens during the 'elution from the column' phase in chromatography?

- a) Components with greatest affinity elute first
- ☒ b) Components with least affinity elute first
- c) Components elute in a random manner
- d) Components elute according to their concentration in the mixture

16. In gel permeation chromatography separation is based on....

- ☒ 1 Size of molecule
- 2 Solubility
- 3 Ion exchanging
- 4 Affinity

17. What is the color of Phenolphthalein in basic solutions :

- ☒ 1 Pink
- 2 Blue
- 3 Colorless
- 4 Yellow

18. In a titration experiment how do we know the end point has reached?

- ☒ 1 Using indicators
- 2 With an acid
- 3 With a base
- 4 Measuring pH



- 19 What is the color of Phenolphthalein in acidic solutions?
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 - 4 Yellow
- 20 All the following is true regarding the basic principle of spectrophotometer except ?
- 1 is one of the most useful qualitative analysis
 - 2 It depends on how much a chemical substance absorbs light
 - 3 It depends on how much a substance transmits light
 - 4 Its done over a certain range of wavelength
- 21 Amax is defined as:
- 1 coloured compound absorbs maximum light at a particular wavelength
 - 2 relationship between absorbance and concentration
 - 3 explains the transmittance of light
 - 4 it is useful for colorless compounds
- 22 The mobile phase in chromatography is :
- 1 Liquid-gas
 - 2 Solid-gas
 - 3 Liquid-solid
 - 4 Solid-solid
- 23 Flame Photometry is for determining :
- 1 Concentration of metal ions
 - 2 Concentration of colored solutions
 - 3 Concentration of liquids
 - 4 Concentration of organic solutions

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Which of the following is used in inorganic chemical analysis ?

- 1 Flame Photometry
- 2 Chromatography
- 3 Spectrophotometry ✓
- 4 Colorimetry

25 A positive Benedict's test is not given by

- 1 Sucrose
- 2 Lactose
- 3 Maltose ✓
- 4 Glucose

فرمت پاسخ نامه سلول و ملکول و بیوشیمی عملی

251	201	151	101	51	2	1
252	202	152	102	52	1	2
253	203	153	103	53	3	3
254	204	154	104	54	4	4
255	205	155	105	55	4	5
256	206	156	106	56	3	6
257	207	157	107	57	2	7
258	208	158	108	58	4	8
259	209	159	109	59	3	9
260	210	160	110	60	1	10
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262	212	162	112	62	1	12
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293	243	193	143	93		43
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296	246	196	146	96		46
297	247	197	147	97		47
298	248	198	148	98		48
299	249	199	149	99		49
300	250	200	150	100		50